

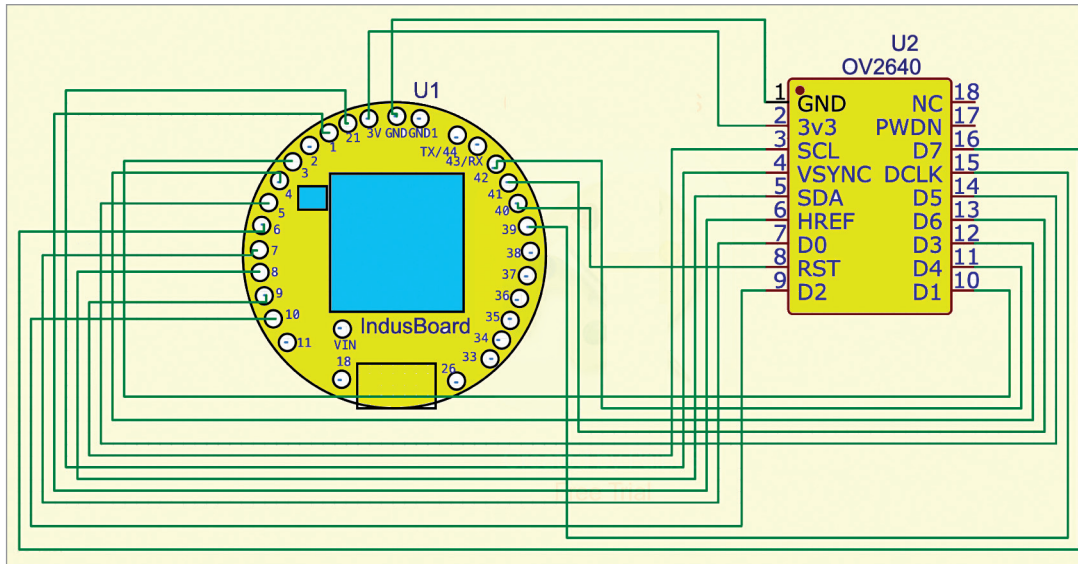


Fig. 3: The circuit diagram

footprint—was essential. The IndusBoard Coin V2, just 3.3cm across (approximately coin-sized), was chosen for its compact design and built-in sensors. The Bill of Materials table lists the components used; Fig. 2 shows the assembled prototype.

Circuit diagram

Fig. 3 illustrates the wiring between the IndusBoard Coin and the OV2640 camera module. Key GPIO pin connections include:

XCLK (GPIO 44): Supplies a 20MHz clock signal

PCLK (GPIO 39): Synchronises pixel data

VSYNC (GPIO 21): Indicates the start of a new frame

HREF (GPIO 1): Denotes the beginning of each line

Y2 to Y9 (GPIOs 7, 3, 10, 4, 42, 5, 41, 6): Carry 8-bit pixel data

SDA (GPIO 8) and SCL (GPIO 9): I²C communication lines for camera setup

tion, a browser-based TensorFlow.js application using the COCO-SSD model overlays bounding boxes on detected objects such as people, animals, and vehicles.

Techy bits: pins, DVP, and PSRAM

The OV2640 camera communicates via a digital video port (DVP), transmitting image data using eight parallel lines along with synchronisation signals. This interface

RESET (GPIO 2): Resets the camera module when required

After the connections are established, the IndusBoard functions as a Wi-Fi server. It captures frames from the OV2640, compresses them as JPEGs, and streams the video feed to a connected phone or computer. For real-time object detec-

```
amobjectdetetion.ino
1  #include "esp_camera.h"
2  #include <WiFi.h>
3  #include <WebServer.h>
4
5  #define PWDN_GPIO_NUM    -1
6  #define RESET_GPIO_NUM   2
7  #define XCLK_GPIO_NUM    44
8  #define SIOD_GPIO_NUM    8
9  #define SIOC_GPIO_NUM    9
10 #define Y9_GPIO_NUM       6
11 #define Y8_GPIO_NUM      41
12 #define Y7_GPIO_NUM       5
13 #define Y6_GPIO_NUM      42
14 #define Y5_GPIO_NUM       4
15 #define Y4_GPIO_NUM      10
16 #define Y3_GPIO_NUM       3
17 #define Y2_GPIO_NUM       7
18 #define VSYNC_GPIO_NUM   21
19 #define HREF_GPIO_NUM     1
20 #define PCLK_GPIO_NUM    39
21
```

Fig. 4: Snippet of source code showing pin configuration and library inclusion

```
8  #define SIOD_GPIO_NUM    8
9  #define SIOC_GPIO_NUM    9
10 #define Y9_GPIO_NUM       6
11 #define Y8_GPIO_NUM      41
12 #define Y7_GPIO_NUM       5
13 #define Y6_GPIO_NUM      42
14 #define Y5_GPIO_NUM       4
15 #define Y4_GPIO_NUM      10
16 #define Y3_GPIO_NUM       3
17 #define Y2_GPIO_NUM       7
18 #define VSYNC_GPIO_NUM   21
19 #define HREF_GPIO_NUM     1
20 #define PCLK_GPIO_NUM    39
21
22 const char* ssid = "JioFiber-4G"; // Replace with your WiFi SSID
23 const char* password = "1234567890"; // Replace with your WiFi password
24
25 WebServer server(80);
26 framesize_t frameSize = FRAMESIZE_VGA; // Higher resolution for better detection
27 pixformat_t pixelFormat = PIXFORMAT_JPEG;
28
29 void initCamera() {
30     camera_config_t config;
31     config.ledc_channel = LEDC_CHANNEL_0;

```

Fig. 5: Wi-Fi SSID and password input section in the code